

Introdução à análise de redes Bayesianas

Estrutura:

1. Introdução
2. Conceitos básicos
3. Estimação de parâmetros e inferência
4. Construção de redes Bayesianas orientadas por dados

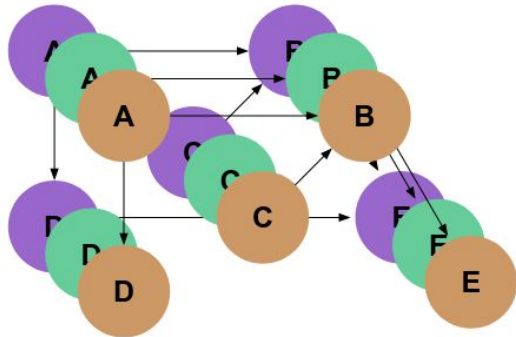
Extra:

- 5. Rede média: bootstrapping**

Rede média: bootstrapping

Rede média: bootstrapping

A	B	C	D	E		
1	A	B	C	D	E	
0	1	A	B	C	D	E
1	1	0	0	1	1	0
...	0	1	0	0	1	0
...	1	1	0	1	0	
		...	...	...	...	...

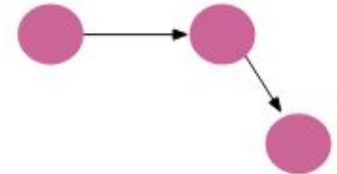
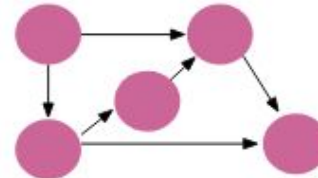
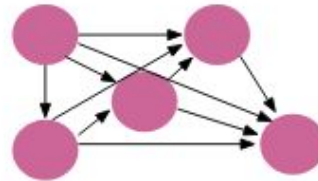
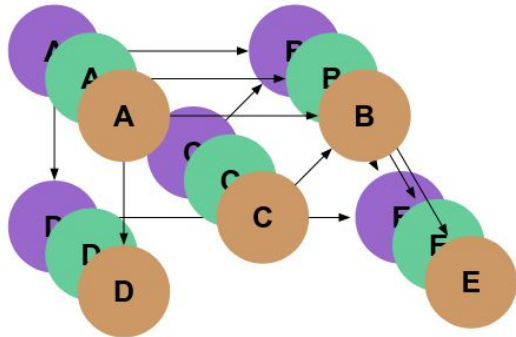


Rede média: bootstrapping

A	B	C	D	E		
1	A	B	C	D	E	
0	1	A	B	C	D	E
1	1	0	0	1	1	0
	0	1	0	0	1	0
...		1	1	0	1	0
		...	...	...	...	...

Arestas presentes em
pelo menos uma rede

Arestas presentes
em todas as redes



Rede média: bootstrapping (exemplo)

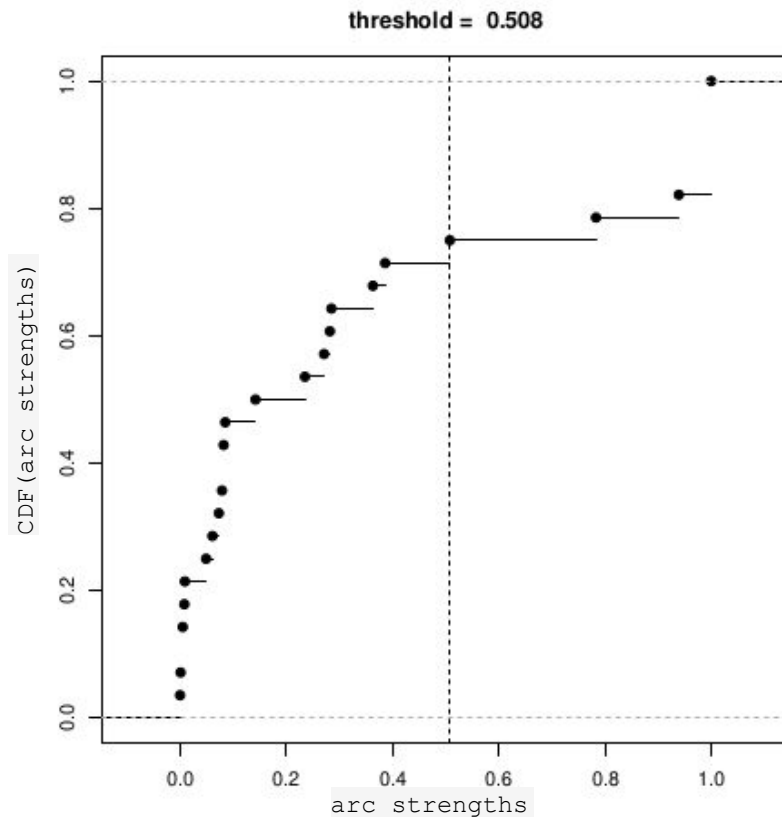
```
boot = boot.strength(data = asia, R=1000, algorithm = 'hc',  
algorithm.args = list(score = 'bde'))
```

```
par(mar=c(3,3,3,3))  
plot(boot)
```

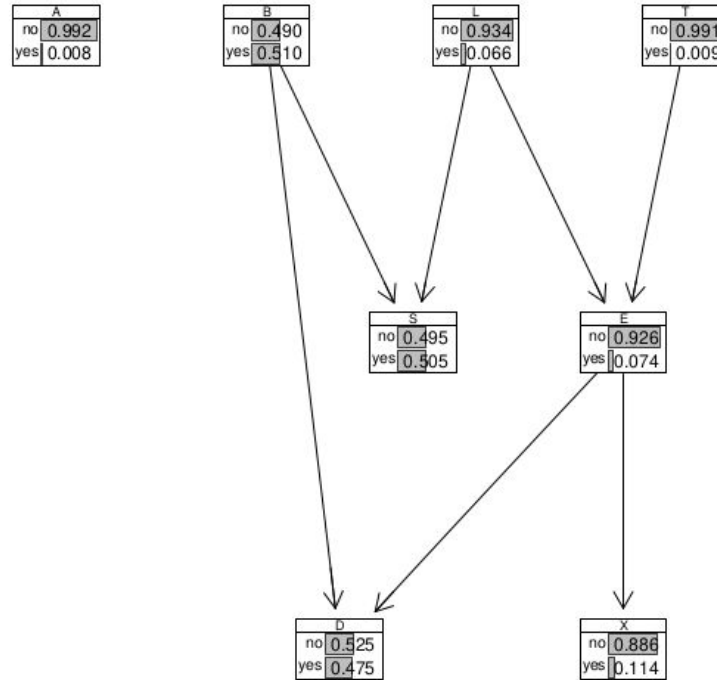
```
boot.avg = averaged.network(boot)  
undirected.arcs(boot.avg)  
graphviz.plot(boot.avg, layout = "dot")
```

```
fit = bn.fit(boot.avg, asia, method ="mle")  
graphviz.chart(fit, type = "barprob")
```

Rede média: bootstrapping (exemplo)



Rede média: bootstrapping (exemplo)



Rede média: bootstrapping (outro limiar)

```
boot.avg = averaged.network(boot, threshold = 0.3)
undirected.arcs(boot.avg)
graphviz.plot(boot.avg, layout = "dot")

fit = bn.fit(boot.avg, asia, method = "mle")
graphviz.chart(fit, type = "barprob")
```

Rede média: bootstrapping (outro limiar)

